

ISTANBUL TECHNICAL UNIVERSITY Department of Electrical Engineering

ENGINEERING DRAWING (ELK 363E)

Orthographic Projection

Assoc. Prof. Dr. Ali Sinan ÇABUK

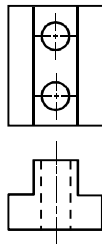
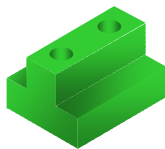
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TOPICS

- Object representation
- Multiview projection
- Glass box concept
- Orthographic projection of **point**, **line**, **plane**, **surface** and **object**.
- Line convention

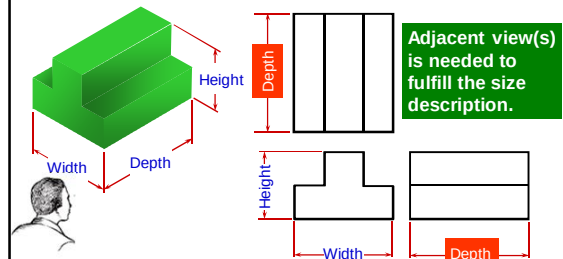
OBJECT REPRESENTATION

- Axonometric projection
- Multiview projection



MULTIVIEW PROJECTION

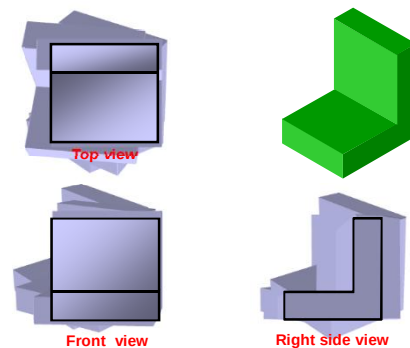
Three principle dimensions of an object ... can be presented only two in each view.

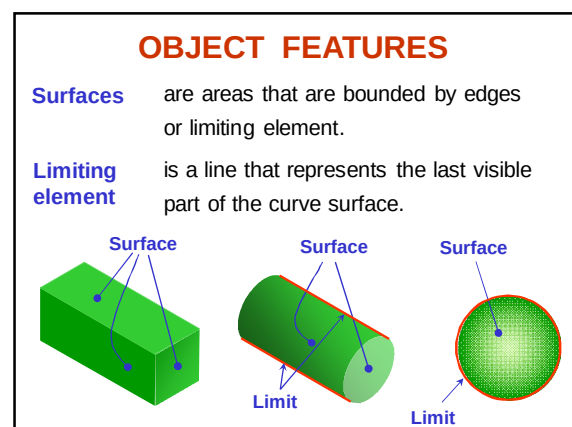
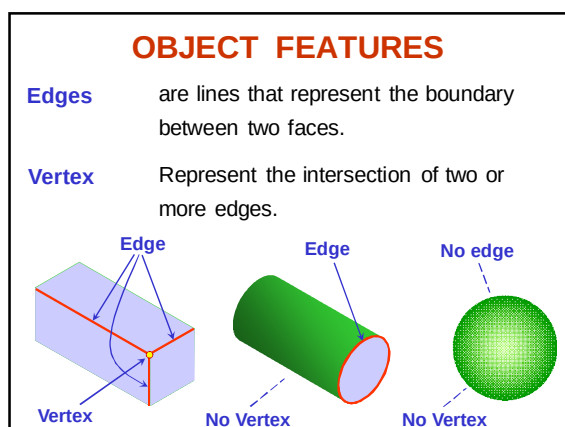
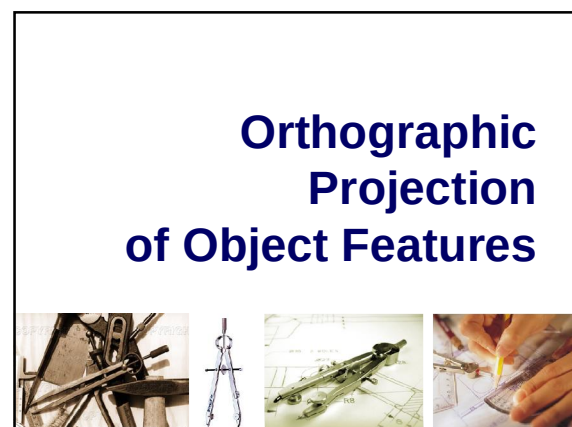
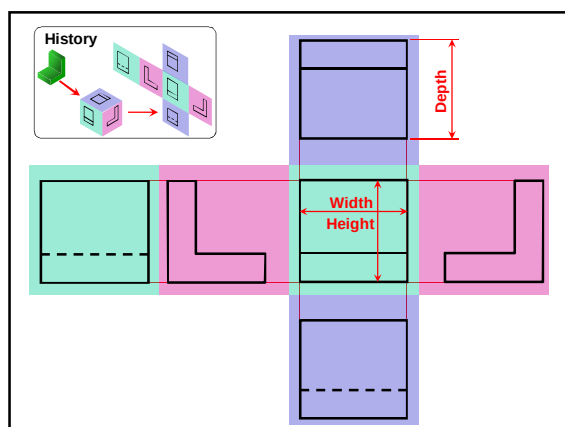
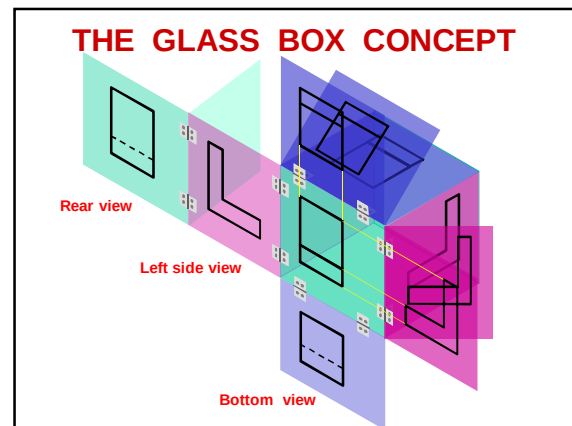
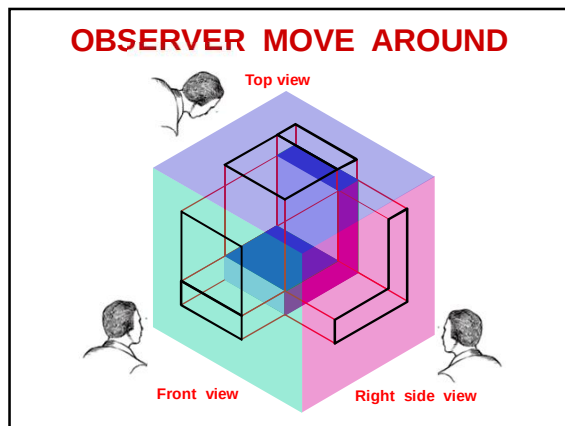


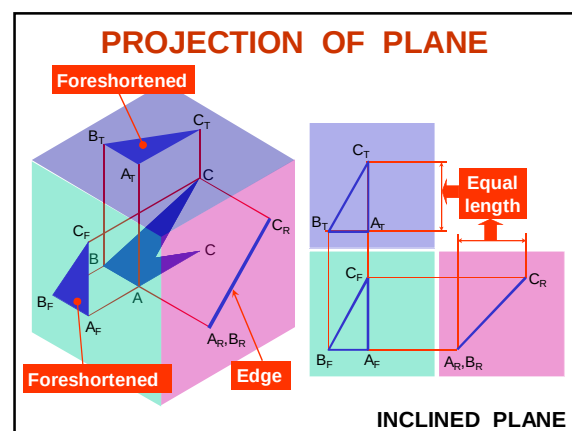
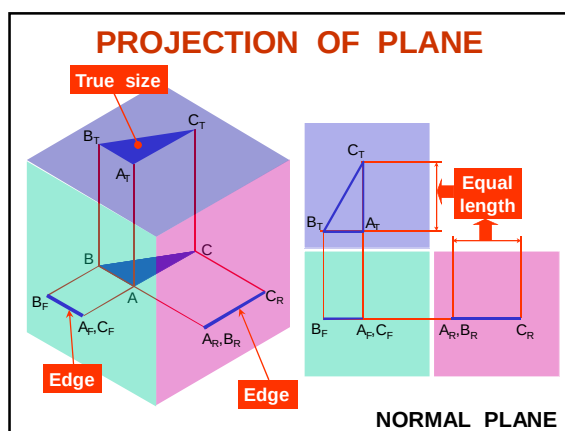
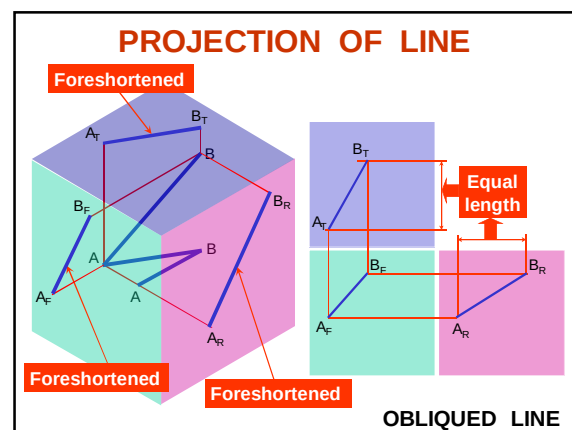
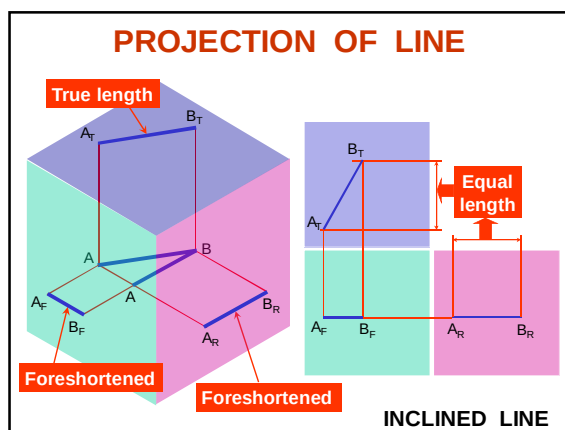
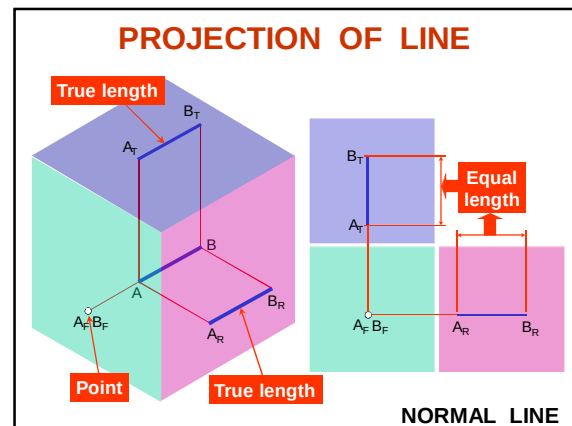
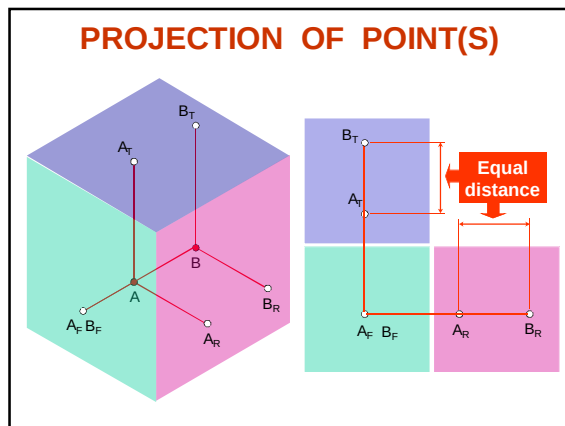
TO OBTAIN MULTIVIEW REPRESENTATION OF AN OBJECT

1. Revolve the **object** with respect to **observer**.
2. The **observer** move around the **object**.

REVOLVE THE OBJECT







PROJECTION OF PLANE

The diagram illustrates the projection of a plane, showing the relationship between the object, its projections, and the equal length dimension.

Object: A 3D object is shown in the top left, with vertices labeled $A, B, C, A_T, B_T, C_T, A_F, B_F, C_F, A_R, B_R, C_R$. The object is a triangular prism-like shape with a base triangle ABC and a top triangle $A_T B_T C_T$. The front face is $A_F B_F C_F$, the top face is $A_T B_T C_T$, and the right face is $A_R B_R C_R$.

Projections: The object is projected onto three planes:

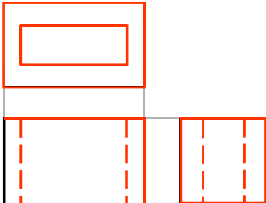
- Top View (Horizontal Plane):** The projection of the object onto the horizontal plane, showing the true shape and size of the base triangle ABC . The vertices are labeled A, B, C .
- Front View (Vertical Plane):** The projection of the object onto the vertical plane, showing the true shape and size of the front face $A_F B_F C_F$. The vertices are labeled A_F, B_F, C_F .
- Right View (Profile Plane):** The projection of the object onto the profile plane, showing the true shape and size of the right face $A_R B_R C_R$. The vertices are labeled A_R, B_R, C_R .

Equal Length Dimension: A red arrow labeled "Equal length" indicates that the length of the object (the distance between A and B) is equal to the length of its projections on the horizontal and profile planes ($A_F B_F$ and $A_R B_R$).

OBLIQUE PLANE

PROJECTION OF OBJECT

The views are obtained by projecting all object features to the picture plane.



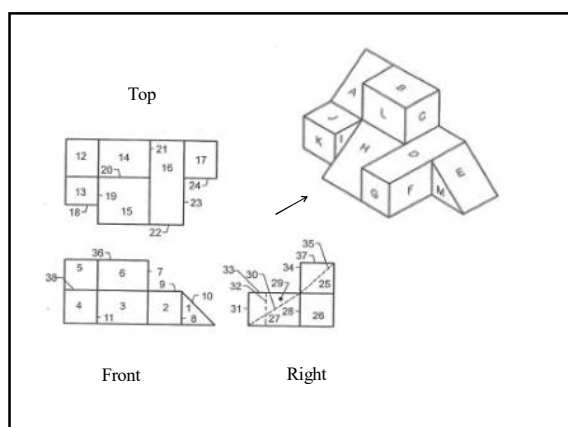
You have to project the remaining surfaces which are invisible too !

PROJECTION OF OBJECT

The diagram illustrates the orthographic projection of a 3D object, a quarter-cylinder block, onto two planes. The object is shown in isometric view on the left, colored red. It consists of a rectangular base with a quarter-cylinder on top. The orthographic projections are shown on the right. The front view (top) is a rectangle with a height dimension 'S' indicated by a red arrow. The side view (bottom) is a rectangle with a width dimension 'S' indicated by a red arrow. The top view (left) shows a quarter-circle arc.

PROJECTION OF OBJECT

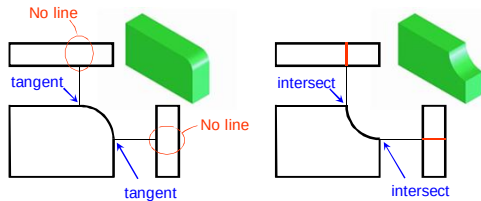
The diagram illustrates the orthographic projection of a 3D object. On the left is the isometric view of a red L-shaped block with a semi-circular cutout on its top surface. On the right are the three standard orthographic views: the front view (top), the top view (bottom), and the side view (right). The front view shows a rectangle with a semi-circular cutout at the bottom. The top view shows a rectangle with a semi-circular cutout at the top. The side view shows a rectangle with a semi-circular cutout at the top. The object is colored red, and the views are outlined in black.



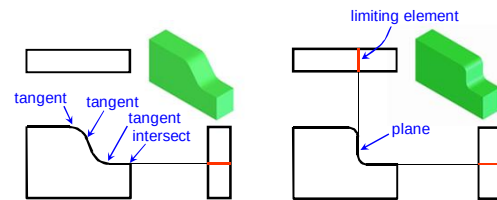
A collage of four images. The top-left image shows a red wall. The top-middle image is a floor plan with labels like 'DINING ROOM' and 'KITCHEN'. The top-right image shows a woman smiling and holding a folder. The bottom-right image shows a person sitting at a desk, looking at a computer screen, with a filing cabinet in the background.

TANGENT & INTERSECTION

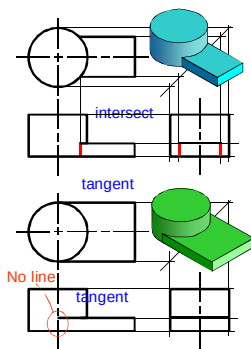
- No line is formed when curved surface **tangent** to a plane surface.
- Line is formed when curved surface **intersects** a plane surface



TANGENT & INTERSECTION



TANGENT & INTERSECTION



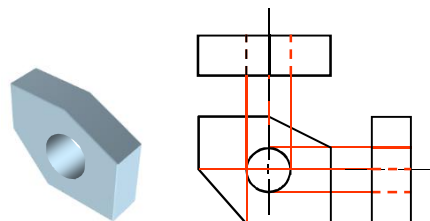
Line Convention

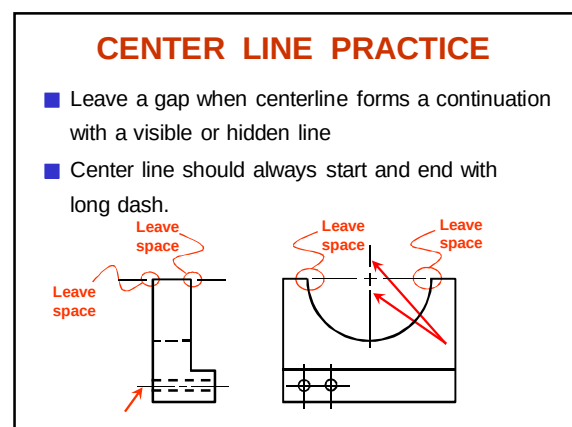
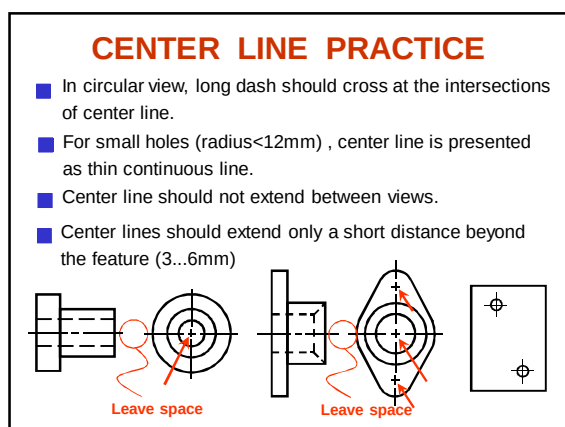
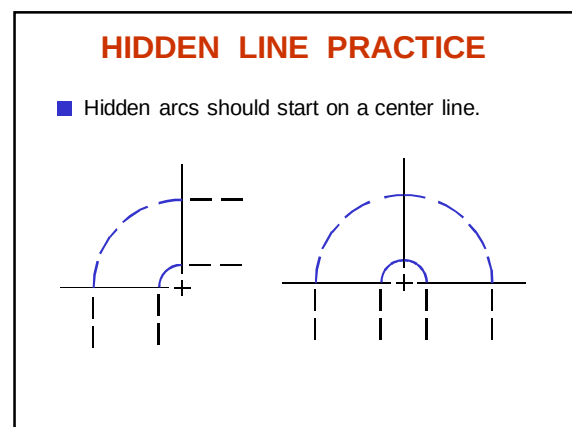
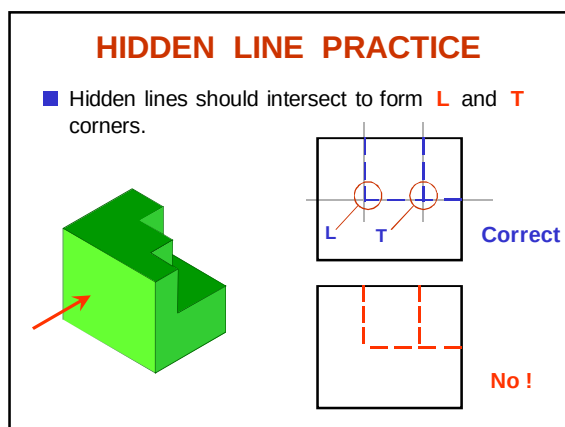
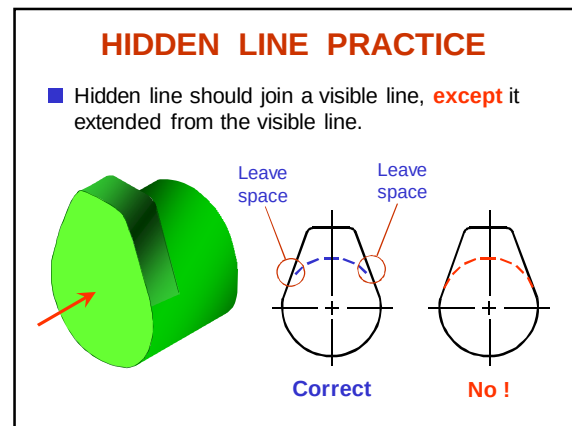
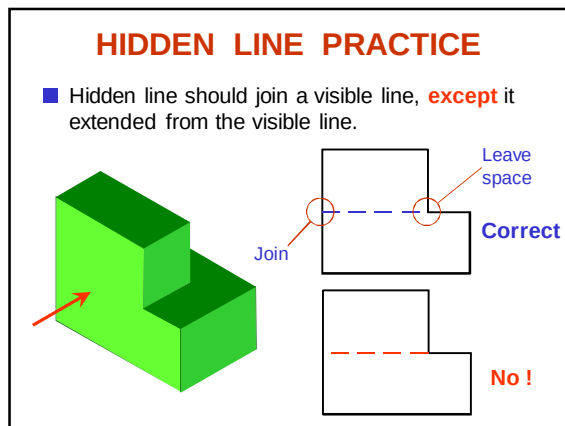
LINE CONVENTION

- Precedence of coincident lines.
- Hidden line drawing.
- Center line drawing.

PRECEDENCE OF LINES

Order of importance: Visible line → Hidden line → Center line





	CORRECT	INCORRECT
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

General rules on line work

- 1 All chain lines should start and finish with a long dash.
- 2 Centre lines should cross one another at long-dash portions of the line at the centres of circles, arcs, etc.
- 3 Centre lines should extend only a short distance beyond the feature, unless required for dimensioning etc.
- 4 A centre line should not extend through the spaces between views.
- 5 A centre line should not terminate at another line.
- 6 Where angles are formed in chain lines, long dashes should meet or cross at the intersections.
- 7 Arcs should join at tangent points.
- 8 Dashed lines should start and end with dashes in contact with the hidden or visible outline.
- 9 Dashed lines should meet or cross with dashes at the intersection.
- 10 If a dashed line meets a curved line tangentially, it should be with solid portions of the lines.